

or ***Felis leo* L.** It indicates that this species was first described by Linnaeus. *The year of the discovery is written after the name of the person who discovered it - e.g. **Felis leo Linnaeus, 1758**. When the name of the genus is not the one under which the original author placed a species, or if the generic name is changed subsequently, the original author's name and the year are kept in parenthesis e.g. **Panthera leo (Linnaeus, 1758)**. It is written so, to understand that Linnaeus originally placed the species name 'leo' under the genus 'Felis' and it was later shifted to the genus Panthera.*

II. Trinominal Nomenclature

It is the extension of the binominal system of nomenclature. This system permits the designation of subspecies with a three-worded name called '**trinomen**'. This system of nomenclature is used to name the subspecies, which is a category below the level of species. The word denoting the name of sub species also begins with a small letter and it is a latinised word, printed in italics e.g. ***Homo sapiens sapiens***; ***Corvus splendens splendens***.

TAUTONYMY: The practice of naming the animals, in which the generic name and species name are the same, is called tautonymy. So the name is called tautonym e.g. ***Axis axis*** (spotted deer); ***Naja naja*** (the Indian cobra).

1.8 SPECIES CONCEPT

I. Species

Species is the 'basic unit' of classification. Species is a Latin word meaning 'kind' or 'appearance'. John Ray in his book 'Historia Generalis Plantarum', used the term 'species' and described it on the basis of common descent (origin from common ancestors) as a group of morphologically similar organisms. Linnaeus considered species, in his book 'Systema Naturae', as the basic unit of classification. Buffon, in his book 'Natural History', proposed the idea of evolution of species which is the foundation for the biological concept of evolution. This biological concept of species (dynamic nature of species) became more popular with the publication of the book "The Origin of Species" by Charles Darwin.

Buffon's biological concept of species explains that species is an interbreeding group of similar individuals sharing the common 'gene pool', and producing fertile offspring. Species is considered a group of individuals which are:

1. Reproductively isolated from the individuals of other species – **a breeding unit.**
2. Sharing the same ecological niche – **an ecological unit.**
3. Showing similarity in the karyotype – **a genetic unit.**
4. Having similar structure and functional characteristics – **an evolutionary unit.**